

PROJECT DESCRIPTION DOCUMENT

Course: BS Computer Science (6th Semester) – Artificial Intelligence

Project Title

AI-Based Long-Term Investment Advisory & Portfolio Optimization System

Duration

4 Weeks

Team Size

3 Students

Project Type

AI-Based Decision Support System (MVP)

1. Project Overview

The goal of this project is to design and implement an intelligent investment advisory system that assists long-term investors in making portfolio allocation decisions based on investment goals, risk appetite, expected return, and investment duration.

The system should use Artificial Intelligence concepts such as:

- heuristic evaluation,
- local search,

- simulated annealing,
- optimization,
- constraint handling,
- explainable recommendations.

The system is NOT expected to perform real-time trading or guarantee future profits. Instead, it should function as an intelligent decision-support and portfolio optimization platform.

2. Example User Queries

The system should support queries such as:

- “I want to invest 5 million PKR for 5 years with medium risk.”
 - “Suggest a diversified portfolio with expected return above 30%.”
 - “Avoid high-risk technology stocks.”
 - “I prefer stable dividend-paying companies.”
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3. Project Objectives

Students should be able to:

- model investment decisions as an AI optimization problem,
 - define heuristic scoring systems,
 - implement portfolio optimization algorithms,
 - apply local search and simulated annealing,
 - compare optimization strategies,
 - generate explainable investment recommendations,
 - visualize investment allocations and risk-return tradeoffs.
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4. Functional Requirements

The MVP must include the following components.

4.1 Investor Profile Module

The system should collect:

- investment amount,
- investment duration,
- risk appetite,
- target return,
- preferred sectors,
- excluded sectors.

Example:

Parameter	Example
Investment Amount	5,000,000 PKR
Duration	5 Years
Risk Appetite	Medium
Target Return	30%
Preferred Sector	Banking
Excluded Sector	Crypto

4.2 Stock Knowledge Base

The system should maintain stock-related information such as:

- company name,
 - sector,
 - historical growth,
 - volatility,
 - dividend yield,
 - stability score,
 - risk score.
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4.3 AI Recommendation Engine

The system should compute stock scores using heuristic evaluation.

Example heuristic factors:

Factor	Weight
Long-Term Growth	+30
Low Volatility	+20
Dividend Stability	+15
Sector Diversification	+20
Momentum Trend	+15

Students may design their own scoring strategy.

4.4 Portfolio Optimization Module

The system must optimize portfolio allocation using:

Required:

- Hill Climbing

Required:

- Simulated Annealing

Students must compare both methods experimentally.

Optimization goals may include:

- maximizing expected return,
 - minimizing risk,
 - improving diversification,
 - satisfying constraints.
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4.5 Explainability Module

The system must explain recommendations.

Example:

“MCB was selected because of:

- strong historical stability,
 - low volatility,
 - consistent dividend yield.”
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4.6 Visualization Dashboard

The system must provide graphical outputs such as:

- portfolio allocation pie chart,
 - sector distribution,
 - risk vs return graph,
 - projected investment growth,
 - comparison of optimization methods.
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5. AI Concepts Expected

The project should demonstrate understanding of the following AI concepts:

AI Topic	Expected Usage
Heuristics	Stock evaluation
Local Search	Portfolio optimization
Hill Climbing	Initial optimization
Simulated Annealing	Escaping local optima
CSP/Constraints	Budget/risk restrictions
Utility Functions	Risk-return balancing
Explainable AI	Recommendation reasoning

6. Recommended Resources

6.1 Recommended Datasets

Students may use:

Pakistan Stock Exchange (PSX)

- Historical stock CSV data

Yahoo Finance

- Downloadable historical stock data

Kaggle Financial Datasets

Examples:

- stock historical prices,
 - dividend datasets,
 - market sector datasets.
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6.2 Suggested Stock Features

Feature	Description
Avg Annual Growth	Long-term return trend
Volatility	Risk level
Dividend Yield	Passive income potential
Sector	Diversification
Historical Stability	Long-term consistency

6.3 Recommended Tools & Technologies

Programming Language

- Python
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Recommended Libraries

Purpose	Suggested Tools
Data Processing	pandas, numpy
Visualization	matplotlib, plotly
Dashboard/UI	streamlit or flask
Optimization	custom implementation
CSV Handling	pandas

Optional

- yfinance API
- seaborn

- scikit-learn (optional only)
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7. Important Constraints

Students are NOT required to:

- implement real-time trading,
- connect to brokerage APIs,
- use deep learning,
- predict exact future prices,
- build enterprise-scale systems.

The focus is: # intelligent recommendation and optimization.

8. Team Task Distribution Guidelines

Student 1 — Data & Knowledge Engineer

Responsibilities:

- stock data collection,
- preprocessing,
- feature extraction,
- stock scoring engine,
- knowledge base management.

Deliverables:

- cleaned dataset,
 - scoring metrics,
 - heuristic documentation.
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Student 2 — AI Optimization Engineer

Responsibilities:

- hill climbing implementation,
- simulated annealing implementation,
- portfolio optimization,
- experimental comparison.

Deliverables:

- optimization algorithms,
 - comparison analysis,
 - optimization reports.
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Student 3 — Interface & Visualization Engineer

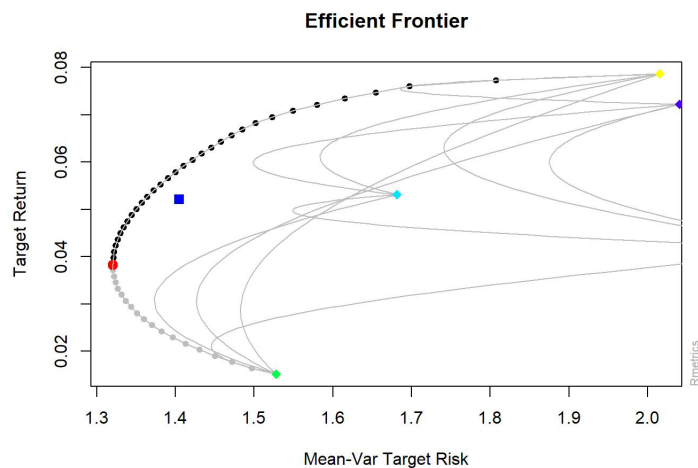
Responsibilities:

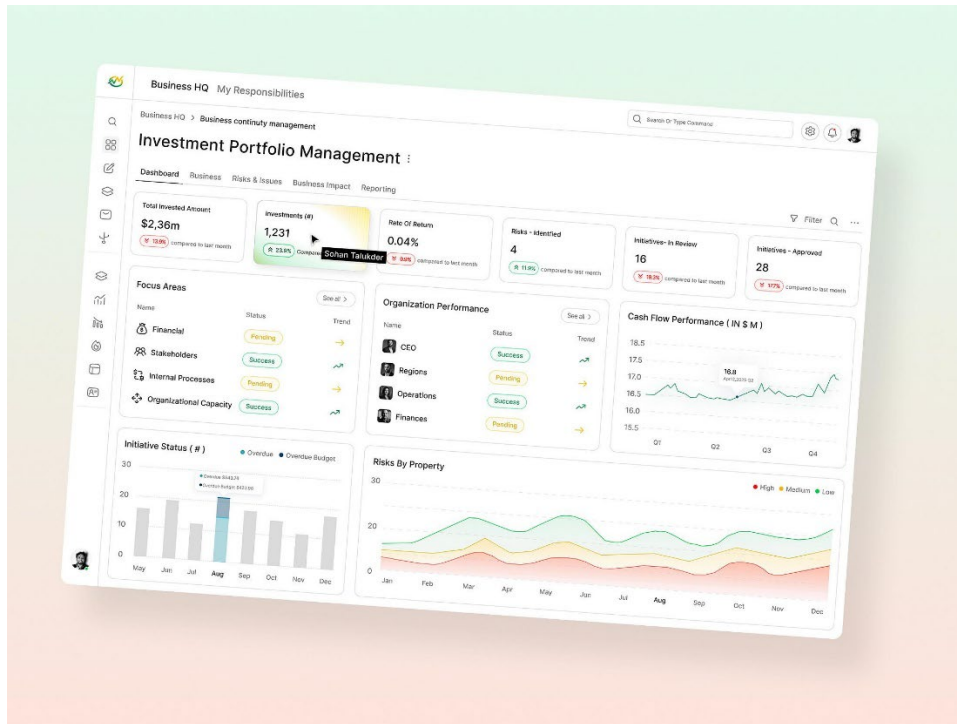
- user interface,
- dashboard,
- visualization,
- explainability module,
- report integration.

Deliverables:

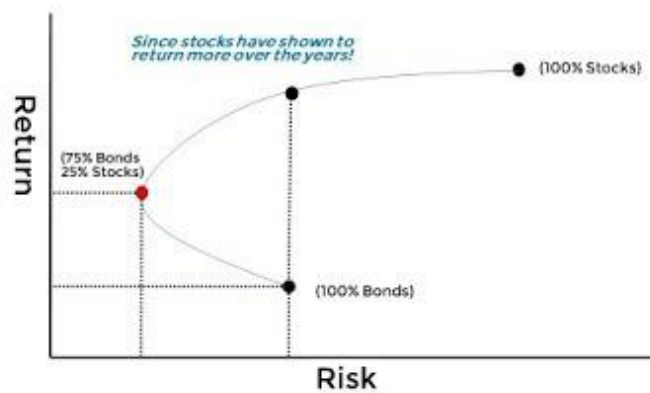
- graphs/charts,
- recommendation interface,
- explainability reports.

Sample Interactive Outputs:





Efficient Frontier



9. Weekly Deliverables & Evaluation

Week 1 — Problem Modeling & Data Preparation

Weight: 20%

Deliverables

- finalized problem definition,
- dataset collection,
- stock features identified,
- initial UI mockup,
- heuristic design draft.

Evaluation Criteria

- clarity of modeling,
 - dataset quality,
 - feasibility of design.
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Week 2 — AI Recommendation Engine

Weight: 25%

Deliverables

- stock scoring system,
- investor profile module,
- initial recommendation logic,
- basic visualization.

Evaluation Criteria

- quality of heuristics,
 - explainability,
 - recommendation reasoning.
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Week 3 — Optimization Algorithms

Weight: 30%

Deliverables

- hill climbing implementation,

- simulated annealing implementation,
- portfolio optimization results,
- comparison experiments.

Evaluation Criteria

- correctness of optimization,
 - algorithm comparison,
 - handling of constraints.
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Week 4 — Final Integration & Demonstration

Weight: 25%

Deliverables

- complete working MVP,
- final report,
- presentation,
- oral defense,
- demonstration.

Evaluation Criteria

- system integration,
 - AI understanding,
 - visualization quality,
 - explainability,
 - oral performance.
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10. Oral Examination Guidelines

Each student will be examined individually.

Students must explain:

- system architecture,
- heuristic design,
- optimization strategy,
- simulated annealing behavior,
- risk-return tradeoffs,

- explainability logic.
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Example Oral Questions

- Why did your system recommend this portfolio?
 - Why is simulated annealing useful here?
 - Why can hill climbing get stuck?
 - How did you model investment risk?
 - What are the constraints in your optimization problem?
 - How does your scoring function work?
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11. Final Deliverables

Each team must submit:

- source code,
 - dataset,
 - final report,
 - presentation slides,
 - demonstration video (optional),
 - optimization comparison report.
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12. Bonus Features (Optional)

Additional marks may be awarded for:

- advanced visualizations,
 - sentiment analysis,
 - multi-agent investor simulation,
 - dynamic market simulation,
 - interactive dashboards,
 - advanced explainability.
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13. Expected Learning Outcomes

After completing the project, students should be able to:

- formulate real-world problems as AI optimization problems,
 - apply heuristic reasoning,
 - implement local search algorithms,
 - understand optimization tradeoffs,
 - design explainable AI systems,
 - build practical AI-driven decision support systems.
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